Comparing Matrix Multiplication Implementations Across CPU and GPU Platforms

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Abstract

1 Introduction

2 Implementations

3 Experimental Method

3.1 Hardware and software

3.2 Workload and correctness

3.3 Timing procedure

3.4 Precision

4 Results

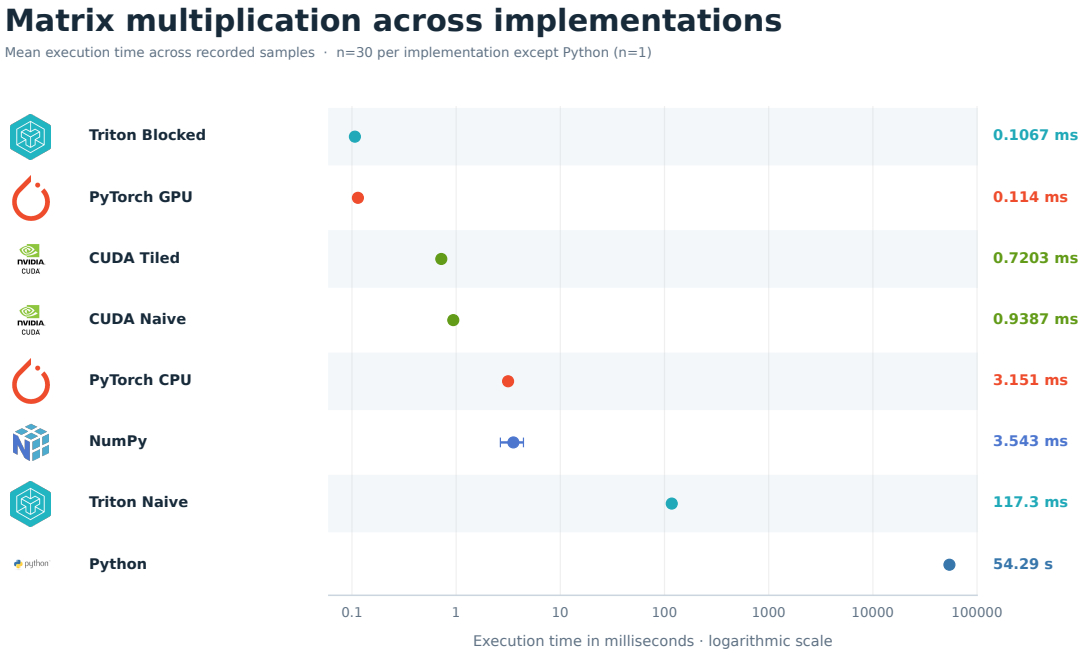


Figure 1: Mean execution time across implementations. The horizontal axis is logarithmic. Python has one measured sample; other implementations have 30.

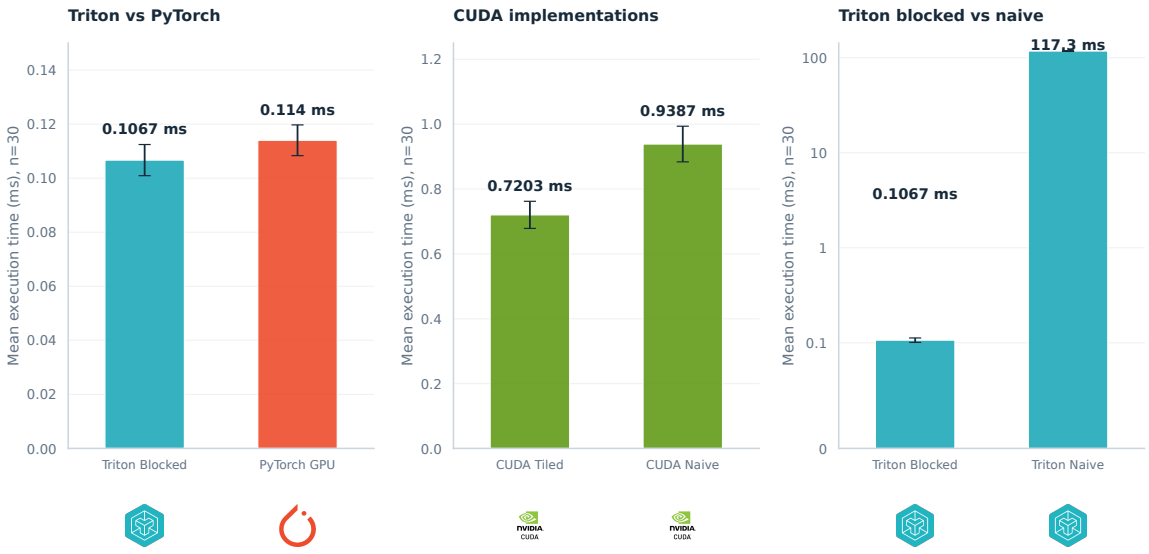


Figure 2: GPU execution time comparisons. The Triton blocked-versus-naive panel uses a symmetric logarithmic vertical scale to show both timings.



Figure 3: Individual measured timing samples and their distributions. Each panel uses its own horizontal scale.

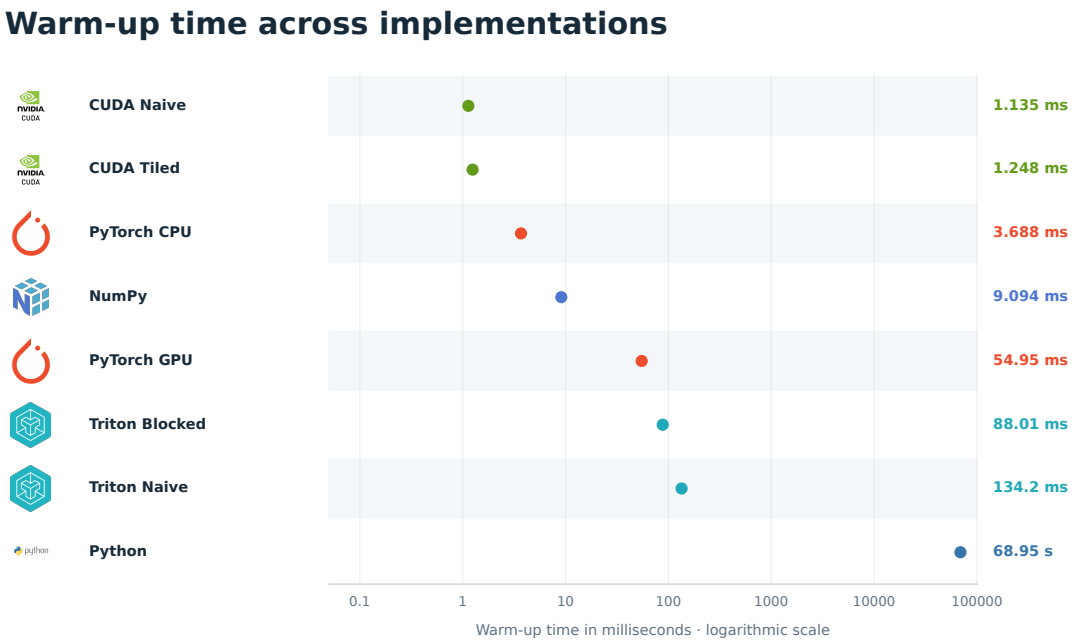


Figure 4: Separately recorded warm-up times. These are not included in the execution-time sample statistics.

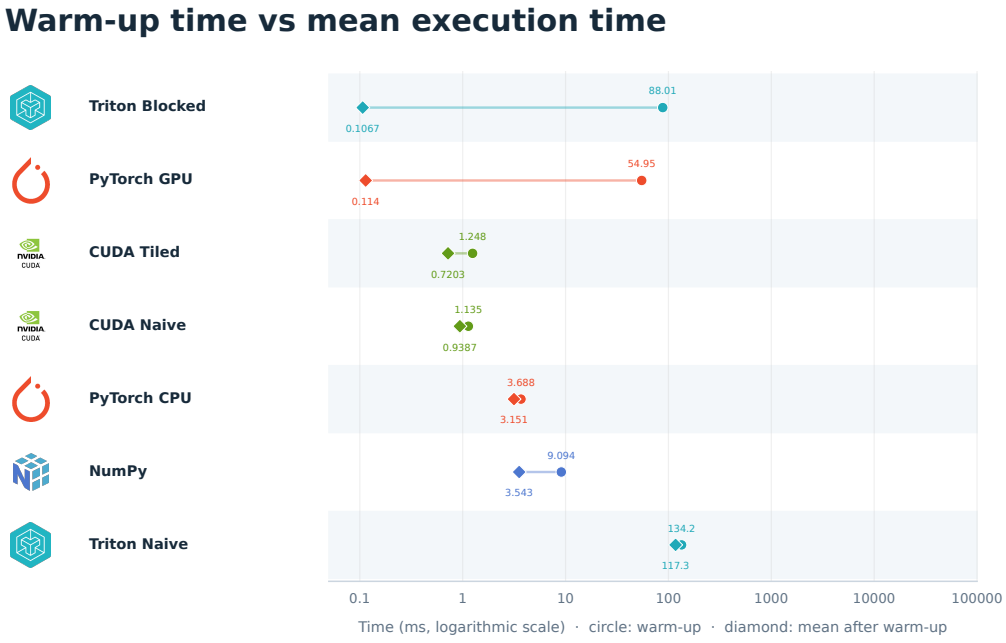


Figure 5: Warm-up time and mean measured execution time for implementations with repeated samples. Circles show warm-ups; diamonds show measured means.

5 Discussion

6 Limitations

7 Conclusion

Reproducibility

References

[1] OpenAI, *Triton documentation*, https://triton-lang.org/main/.

[2] PyTorch contributors, *PyTorch documentation*, https://pytorch.org/docs/stable/.

[3] NVIDIA, *CUDA C++ Programming Guide*, https://docs.nvidia.com/cuda/cuda-c-programming-guide/.